3.5 Pre-Quiz: Trigonometry Review — Markscheme

1. [Maximum mark: 11]

$\cos\theta = \frac{1}{4}, \quad \frac{3\pi}{2} < \theta < 2\pi$ (fourth quadrant).

(a) $\sin^2\theta = 1 - \left(\frac{1}{4}\right)^2 = \frac{15}{16}$ **M1**

$\sin\theta = \pm\frac{\sqrt{15}}{4}$ **A1**

$\sin\theta = -\frac{\sqrt{15}}{4}$, since θ is in the fourth quadrant **A1**

Accept a right-triangle method (sides 1, $\sqrt{15}$, 4*) for the M1A1. The final A1 is for the negative sign. Wrong method: the GDC route* $\sin\left(\arccos\frac{1}{4}\right) = +\frac{\sqrt{15}}{4}$ *gives the first-quadrant angle, earning M1A1A0. Decimals (*-0.968*) earn the final A0.*

(b) $\sin 2\theta = 2 \times \left(-\frac{\sqrt{15}}{4}\right) \times \frac{1}{4} = -\frac{\sqrt{15}}{8}$ **M1A1**

FT from (a): $+\frac{\sqrt{15}}{4}$ *in (a) gives* $+\frac{\sqrt{15}}{8}$*, M1A1 FT. Wrong method:* $\sin 2\theta = 2\sin\theta = -\frac{\sqrt{15}}{2}$ *earns M0A0.*

(c) $\cos 2\theta = 2 \times \left(\frac{1}{4}\right)^2 - 1 = -\frac{7}{8}$ **M1A1**

Accept any of the three forms, e.g. $\frac{1}{16} - \frac{15}{16}$ *or* $1 - 2 \times \frac{15}{16}$*. Wrong method:* $\cos 2\theta = 2\cos\theta = \frac{1}{2}$ *earns M0A0. A value outside* $[-1, 1]$ *signals an error.*

(d) $\frac{\sin 2x}{1 + \cos 2x} = \frac{2\sin x \cos x}{1 + 2\cos^2 x - 1}$ **M1**

$= \frac{2\sin x \cos x}{2\cos^2 x}$ **A1**

$= \frac{\sin x}{\cos x}$ **A1**

$= \tan x$ **AG**

M1 for substituting both double-angle identities. The form $\cos 2x = 2\cos^2 x - 1$ *is the efficient choice;* $\cos^2 x - \sin^2 x$ *also works, with* $1 - \sin^2 x = \cos^2 x$*. Wrong method: checking one value, e.g.* $x = \frac{\pi}{4}$*, proves nothing and earns M0.*

(e) $\tan\theta = \frac{-\frac{\sqrt{15}}{8}}{1 - \frac{7}{8}} = -\sqrt{15}$ **A1**

Or otherwise: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-\sqrt{15}/4}{1/4}$*. FT from (b) and (c), or from (a). Inverted ratio* $-\frac{1}{\sqrt{15}}$ *earns A0.* $+\sqrt{15}$ *earns A0 unless it follows from the candidate's* $+\frac{\sqrt{15}}{4}$ *in (a).*

2. [Maximum mark: 12]

$HA = 12$ km on 070°, then $AB = 9$ km on 170°.

(a) bearing of H from $A = 070° + 180° = 250°$ **M1**

$H\hat{A}B = 250° - 170° = 80°$ **A1 AG**

Accept any correct angle argument at A, e.g. the angle between $[AH]$ and the south line at A is 70° (alternate angles) and the angle between $[AB]$ and the south line is 10°, so $70° + 10° = 80°$; or the ship turns through $170° - 70° = 100°$, so $H\hat{A}B = 180° - 100°$.

(b) $HB^2 = 12^2 + 9^2 - 2 \times 12 \times 9\cos 80°$ **M1A1**

$HB = 13.7$ km $\quad (13.6927\ldots)$ **A1**

M1 for substituting into the cosine rule, A1 for the correct substitution. $HB^2 = 187.491\ldots$. Wrong methods: the angle 100° gives 16.2, M1A0A0; radian mode gives 15.8 (15.7747...), M1A1A0.

(c) $\dfrac{\sin A\hat{H}B}{9} = \dfrac{\sin 80°}{13.6927\ldots}$ **M1**

$A\hat{H}B = 40.3° \quad (40.3380\ldots)$ **A1**

bearing $= 70° + 40.3380\ldots°$ **M1**

$= 110° \quad (110.338\ldots)$ **A1**

Accept 110.3°. Accept the cosine rule for $A\hat{H}B$. The angle is opposite the shorter side $9 < 13.69$, so it is acute: no ambiguous case. FT from (b). Wrong methods: 40.3° given as the bearing, M1A1M0A0; $70° - 40.3° = 29.7°$, M1A1M0A0; the angle at B, 59.7°, added to 70° gives 130°, M0A0M1A0.

(d) $110.338\ldots° + 180° = 290° \quad (290.338\ldots)$ **A1**

Accept 290.3°. FT from (c). Wrong methods, each A0: $360° - 110° = 250°$; $180° - 110° = 70°$.

(e) area $= \dfrac{1}{2} \times 12 \times 9\sin 80°$ **M1**

$= 53.2$ km^2 $\quad (53.1796\ldots)$ **A1**

Accept any two sides with their included angle, e.g. $\frac{1}{2} \times 12 \times 13.6927\ldots \sin 40.3380\ldots°$. Wrong method: an angle not included between the two sides, e.g. $\frac{1}{2} \times 12 \times 13.69 \sin 80° = 80.9$, earns M0A0. Note: $\sin 100° = \sin 80°$, so the wrong angle from (a) gives the correct area; (a) fixes the angle at 80°.

3. [Maximum mark: 11]

(a) $2\sin x\cos x = \sin x$ **M1**

$2\sin x\cos x - \sin x = 0$

$\sin x\,(2\cos x - 1) = 0$ **M1**

$\sin x = 0 \quad$ or $\quad \cos x = \dfrac{1}{2}$

$x = 0,\ \pi,\ 2\pi$ **A1**

$x = \dfrac{\pi}{3},\ \dfrac{5\pi}{3}$ **A1A1**

Second M1 for factorising (or for setting each factor to zero). The A1 for $\sin x = 0$ *needs all three values. Wrong method: dividing by* $\sin x$ *loses* 0, π, 2π *and earns M1M0A0A1A1. Missing the end points* 0 *or* 2π *earns A0 for that line. Decimals (*1.05, 3.14, 5.24*) earn at most one of the three A1.*

(b)(i) $\cos 2x + 3\cos x + 2 = 2\cos^2 x - 1 + 3\cos x + 2$ **M1**

$= 2\cos^2 x + 3\cos x + 1$ **A1 AG**

M1 for choosing the form $\cos 2x = 2\cos^2 x - 1$. *The forms* $\cos^2 x - \sin^2 x$ *or* $1 - 2\sin^2 x$ *earn the M1 only when* $\sin^2 x$ *is then replaced by* $1 - \cos^2 x$.

(b)(ii) $(2\cos x + 1)(\cos x + 1) = 0$ **M1**

$\cos x = -\dfrac{1}{2} \quad$ or $\quad \cos x = -1$ **A1**

$x = \dfrac{2\pi}{3},\ \dfrac{4\pi}{3}$ **A1**

$x = \pi$ **A1**

Accept the quadratic formula, or a substitution $u = \cos x$, *for the M1. Wrong method: a sign error in the factors,* $\cos x = \frac{1}{2}$ *or* 1, *gives* $x = 0, \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi$*: M1A0, then A1A1 FT. Missing* $x = \pi$ *(a single value from* $\cos x = -1$*) is the most common loss. Decimals (*2.09, 3.14, 4.19*) earn at most A1 of the last two.*

4. [Maximum mark: 13]

$f(x) = 6\sin x\cos x + 1, \quad 0 \le x \le \pi.$

(a) $6\sin x\cos x = 3 \times 2\sin x\cos x$ **M1**

$= 3\sin 2x$, so $f(x) = 3\sin 2x + 1$ **A1 AG**

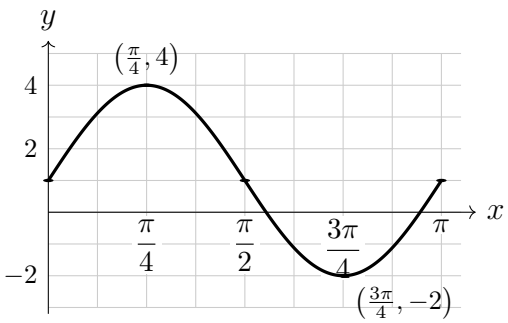
(b) period $= \dfrac{2\pi}{2} = \pi$ **A1**

Wrong methods, each A0: 2π *(the period of* $\sin x$*);* 2.

(c) maximum $= 3 + 1 = 4$, minimum $= -3 + 1 = -2$, so $-2 \le f(x) \le 4$ **A1A1**

A1 for each end point with the correct inequality. Accept $[-2, 4]$*. The end points are included:* $f\left(\frac{\pi}{4}\right) = 4$ *and* $f\left(\frac{3\pi}{4}\right) = -2$*. Strict inequalities earn A1A0. Wrong methods:* $-3 \le f(x) \le 3$ *(the* $+1$ *ignored) or* $-5 \le f(x) \le 7$ *(amplitude* 6*), A0A0.*

(d)



shape: one complete sine wave **A1**

maximum $\left(\frac{\pi}{4}, 4\right)$, minimum $\left(\frac{3\pi}{4}, -2\right)$ **A1**

$y = 1$ at $x = 0, \frac{\pi}{2}, \pi$; domain $[0, \pi]$ **A1**

The x-intercepts are 1.74 *(1.74071...) and* 2.97 *(2.97169...); they need not be labelled. Wrong method: two cycles (period* $\frac{\pi}{2}$*) or half a cycle (period* 2π*) earns A0 for the shape. FT from (b) and (c).*

(e) $3\sin 2x + 1 = 2.5$, so $\sin 2x = \dfrac{1}{2}$ **M1**

$2x = \dfrac{\pi}{6}, \ \dfrac{5\pi}{6} \quad$ (for $0 \le 2x \le 2\pi$)

$x = \dfrac{\pi}{12}, \ \dfrac{5\pi}{12}$ **A1A1**

Wrong methods: $x = \frac{\pi}{6}, \frac{5\pi}{6}$ *(not halved) earns M1A0A0;* $\frac{\pi}{12}$ *only earns M1A1A0;* $\sin 2x = \frac{2.5}{3}$ *(the* $+1$ *ignored) earns M0. Extra values outside* $0 \le x \le \pi$*, e.g.* $\frac{13\pi}{12}$*, earn the final A0. Decimals (*0.262*,* 1.31*) earn A1A0 at most.*

(f) $-2 < k < 1 \quad$ or $\quad 1 < k < 4$ **A1A1**

Accept $-2 < k < 4$*,* $k \ne 1$*. From the sketch:* $k = 1$ *gives three solutions (*0*,* $\frac{\pi}{2}$*,* π*), and* $k = 4$ *or* $k = -2$ *gives one. Wrong methods:* $-2 < k < 4$ *with* $k = 1$ *not excluded earns A1A0; including* $k = -2$ *or* $k = 4$ *earns A0 for that interval.*